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1  def log_func(base,prod):
2      base=float(base)
3      prod=float(prod)
4      exp=1
5      prod_x=base
6      exp_x=0
7      base_x=0
8      exp_add=0
9      exp_state=False
10     if prod<1:
11         exp_state=True
12         prod=1/prod
13     for x in range(10):
14         exp_add=10**-x
15         while prod_x<prod:
16             exp+=exp_add
17             prod_x=base**exp
18         for y in range(10):
19             exp_x=10**-y
20             base_x=base**exp_x
21             if prod_x>prod and exp_add==exp_x:
22                 prod_x=prod_x/base_x
23                 exp=exp-exp_add
24     if exp_state==True:
25         prod_x=1/prod_x
26         exp=-exp
27     return exp
28
29 ln_base=2.718281828459
30
31 def nest_log_func(base,prod):
32     exp_list=[0,1,2]
33     for b,p,e in zip((base,ln_base,ln_base),(prod,base,prod),(exp_list)):
34         exp_list[e]=log_func(b,p)
35     return exp_list
36
37
38 def log_deriv_func(base,prod):
39     global user_exp,log_deriv,log_integ,base_final,prod_final
40     base_final=base
41     prod_final=prod
42     user_exp,ln_exp,ln_exp_int=nest_log_func(base,prod)[0:3]
43     log_deriv=1/(prod*ln_exp)
44     log_integ=(prod/ln_exp)*(ln_exp_int-1)
45     return user_exp,log_deriv,log_integ
46
47 def log_call():
48     print "Ο λογάριθμος της βάσης",base_final,"και του προϊόντος",prod_final,"που
49     δώσατε δίνει εκθέτη",user_exp
50     print "The logarithm of the given base",base_final,"and product",prod_final,
51     "gives exponent",user_exp
52
53 def derivative():
54     print "Το παράγωγο του λογάριθμου είναι",log_deriv
55     print "The derivative of the logarithm is",log_deriv
56
57 def integral():
58     print "Το αόριστο ολοκλήρωμα/αντιπαράγωγο του λογάριθμου, χωρίς σταθερά
59     ολοκληρώσεως είναι",log_integ
60     print "The indefinite integral of the logarithm without the constant of
61     integration is",log_integ
62
63 class log_explain:
64     def calc(self,base,prod):
65         return log_deriv_func(base,prod)
66     def log(self,base,prod):
67         return log_deriv_func(base,prod),log_call()
68     def deriv(self,base,prod):
69         return log_deriv_func(base,prod),derivative()
70     def integ(self,base,prod):
71         return log_deriv_func(base,prod),integral()

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69  a=log_explain()  
70  
71  
72  
73
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